
Breaking Down Context Rot in LLMs

Anonymous Author(s)

Affiliation

Address

email

Abstract

Long conversations are said to wear language models down, but standard evaluations confound accumulation (more context) with dilution (the answer-bearing tokens becoming a smaller, more distant, more contested fraction of it). We separate them in streams of localized tasks, each answerable from its own question, where accumulation grows the context without diluting anything the current answer needs and a span-attention clamp sets any span’s attention share directly. All experiments run on OLMo-2-7B-Instruct.

Accumulation alone costs **nothing**: accuracy and instruction adherence stay flat to 90% fill, with declines larger than 2.9 and 5.5 points excluded at 95%. That null is the paper’s starting point: three other things in a growing context do cost, and length is only the usual way of causing them at once. Displacement, moving a task’s evidence back into the context, costs accuracy, and removing the evidence’s attention mass with nothing else moved reproduces the full penalty, smoothly and without a threshold. Competition, near-duplicate context, costs a further 8.5 points while the evidence’s attention barely moves; closing every context mention of the answer options recovers **56%** against a random-closure control at zero, placing the cost on the competitors rather than the evidence. Precedent, filler that demonstrates an applicable answer shape, erodes an instructed output format completely ($0.875 \rightarrow 0.000$ within three filler turns, accuracy unharmed) while the instruction’s attention holds and its content stays linearly decodable.

The same closure that rescues competition restores nothing for precedent—competition is generation-time reading, while the precedent mode is set at prefill, its carrier localized to the chat template’s delimiter tokens in the lower half of the stack. All of it replicates on Qwen2.5-7B-Instruct except competition (penalty gone, attention signature inverted) and the erosion (staged, not total). Context rot in localized streams is displacement, competition, and precedent, and only the first is attention dilution proper.

1 Introduction

A recurring worry about long conversations is context rot: as turns accumulate, the model is said to degrade. The phenomenology is real. On a repeating multiple-choice task, instruction-tuned models grow more confident as context fills: next-token entropy drops roughly $3\text{--}4\times$ on Qwen-2.5-7B (Yang et al., 2024) and Llama-3.1-8B (Grattafiori et al., 2024). On OLMo-2-Instruct the last token’s attention leaves the system prompt (correlation with fill -0.93) and the current query (-0.71) for the most recent prior cases ($+0.89$), all monotone.

One reading attributes the degradation to length itself. Long context comes in two shapes, and they present different problems. In the first, one task spans the whole context: a long document, a buried needle, a growing codebase. Length there mechanically dilutes the answer-bearing tokens among irrelevant ones, and a measured degradation conflates the cost of accumulation with the cost of the target being harder to attend to (“lost in the middle,” Liu et al., 2023). In the second, the

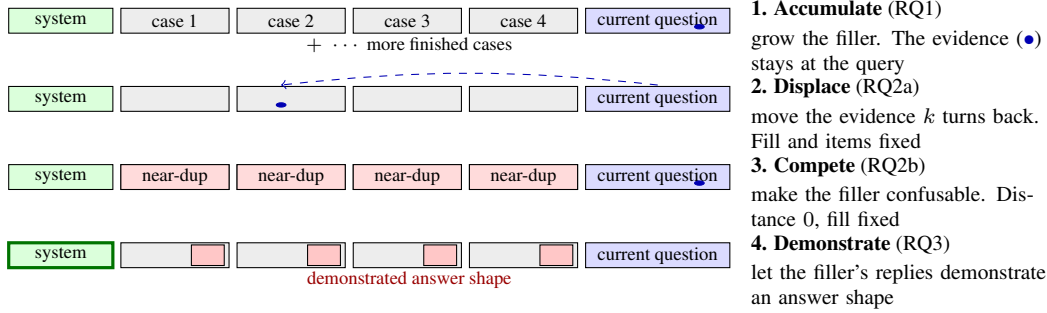


Figure 1: **The localized-stream framework: four independent knobs on one transcript.** Each stream accumulates self-contained cases, so the evidence for the current answer sits at the current question by default and every knob is turned with the others held fixed. The span-attention clamp (§2.3) additionally sets any span’s attention share directly, turning the mediating variable itself.

context is a stream of localized tasks, each self-contained and already finished: an agent transcript of locally-scoped steps, a session of independent questions. Accumulation there grows the context without diluting anything the current answer needs. The context-rot literature measures the first shape almost exclusively, where the two costs cannot be separated. This paper measures the second, where they can and where the two accounts predict opposite outcomes (Appendix A). Retrieval is the case between them, a localized stream whose current answer depends on evidence sitting back in the context, and §4.2 builds it by moving the evidence back.

We separate the candidate causes inside one harness (Fig. 1) and ask three questions. **RQ1**: does accumulation alone, with the evidence for every answer held at the current query, degrade accuracy or instruction adherence? **RQ2**: when dilution does cost accuracy, is the cost mediated by the evidence’s attention mass? We split dilution into its two components, the evidence becoming distant and the context becoming confusable, and test each causally. **RQ3**: if adherence survives context fill, what erodes instruction-following under accumulation?

RQ3 takes up Dongre et al. (2026), who showed that attention to the system prompt closes over accumulated filler and asked whether behavior survives that closure, finding the answer architecture-indexed. Our precedent arms answer with a graded instrument on one architecture.

Accumulation is the design’s null arm (RQ1). Our contributions are the harness and one causal result for each mechanism that costs something:

- A harness that makes accumulation and dilution independent, by accumulating individually localized tasks whose evidence never moves, and a span-attention clamp that sets any span’s post-softmax share with a bit-identical no-op at scale 1 (§2).
- **Displacement**, the retrieval case built inside the stream, is causally sufficient: removing the evidence’s attention mass with nothing moved reproduces the penalty in full, and the mass→accuracy relationship is smooth and graded with no threshold in it (RQ2, §4.2).
- **Competition** is a second mechanism, costing accuracy at constant evidence mass. Closing the competitor mentions recovers 56% of the penalty; both accuracy mechanisms are attentional, on different spans, and span-targeted closure tells them apart (RQ2, §4.3).
- **Precedent** costs compliance rather than accuracy. Accumulated demonstrations erode an instructed format only when their shape transfers to the probe, immediately and with accuracy unharmed. The erosion is not carried by the instruction’s attention, yet re-weighting or re-presenting the instruction each fully reverse it (RQ3, §4.4–§4.5).

2 Methodology

2.1 The localized-stream framework

Our harness accumulates individually localized tasks, the second shape of §1: a stream of self-contained 5-option medical cases (DDXPlus, Tchango et al., 2022) or discrete factual questions (MMLU, Hendrycks et al., 2021). Each answer depends only on the current question, which always sits at the end of the context. Accumulation grows the context without diluting the information the

current answer needs, and each component of dilution becomes an independent variable (Fig. 1). Holding the item, the model, the total fill, and the metric fixed, we can move the evidence k user turns back (displacement, which rebuilds the retrieval case inside the stream), replace the filler with near-duplicate cases (competition), or let the filler’s replies demonstrate a competing answer shape (precedent). Question text is byte-identical across arms, and the filler is identical unless it is the treatment.

2.2 Span share and enrichment

For a token span A (the evidence, the system prompt, a filler answer), its share is the final position’s post-softmax attention mass on A , averaged over heads and pooled over all 32 layers. The ladder and sweep diagnostics (Table 9, Fig. 2) additionally read layer 24 as a reference. A fixed-size span’s raw share falls mechanically as context grows (a 48-token system prompt is a shrinking fraction of the key space), and a raw share decline says nothing by itself. We also report enrichment: share divided by the span’s token fraction, which is 1 for a span attended in proportion to its size.

2.3 The span-attention clamp

To intervene on attention rather than position, we use a clamp that sets a span’s post-softmax share. Adding a constant bias b to a span’s key columns in the additive attention mask multiplies the span’s odds by e^b : a span with unclamped share s at a given head lands at

$$\frac{s_b}{1 - s_b} = e^b \frac{s}{1 - s}, \quad \text{equivalently} \quad s_b = \frac{e^b s}{1 + (e^b - 1)s},$$

with softmax renormalizing the rest of the row. Nothing in the attention forward is reimplemented, and a scale of 1.0 ($b=0$) is bit-identical to the unhooked model. Since s_b is strictly increasing in b , a bisection hits any target share at the reference layer to $\sim 10^{-4}$, and $b \rightarrow -\infty$ (scale 0) removes the span from attention entirely. We call this closing the span. Closure is the hard-masking intervention of Dongre et al. (2026), recovered as the clamp’s endpoint: their ablation and our graded doses sit on one dial. The clamp is bidirectional, and mediation can be tested by restoring mass as well as removing it. It is uniform across heads (per-head structure is measured separately, Appendix F) and, unless stated, applied at every layer. Unlike attention steering aimed at improving compliance (Zhang et al., 2024), the clamp is a measurement instrument with an exact no-op, a solvable dose, and a near-ablation regime excluded from interpretation (§4.2, validation in Appendix B).

2.4 Residual probes and mode-vector steering

To ask what the residual stream represents while behavior changes, we train linear probes (Alain and Bengio, 2016; Belinkov, 2022) on the final pre-generation position across all 33 stack rows, with episode-level leave-one-out cross-validation and permutation nulls. To test what a decodable direction does, we steer with mean-difference vectors at decode time against norm-matched random controls, and erase by projection. Full protocol in Appendix H.

3 Experimental design

Streams and probes. All causal experiments run on OLMo-2-7B-Instruct at seed 42, greedy decoding. Appendix J replicates the program on Qwen2.5-7B-Instruct. Every mechanism experiment runs to a 4k window on both families, and no arm comparison is confounded by window size. The only exceptions are not mechanism results: the adherence arm in §4.1 runs to 32k and the WildChat measurements behind the filler design to 16k and 32k. A 4k window holds only tens of cases; we pool many independent accumulation sessions. A reversed-question-order control removes difficulty-ordering confounds. Probe options are shuffled per item and items with fewer than 5 options are excluded (DDXPlus lists differentials in rank order, and unshuffled gold letters are 71.4% “A”, Appendix C).

Compliance instrumentation. For RQ3 the system prompt carries a clinical answer template (ANSWER: letter, then SUPPORTING: with at least two vignette findings) and every reply is graded separately for compliance and accuracy by a rule-based checker. Simpler canary instructions (a reply

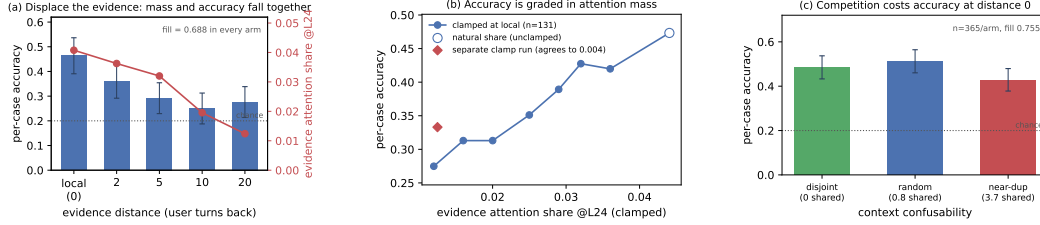


Figure 2: **Dilution costs accuracy. Accumulation does not.** (a) Moving the evidence k user turns back, at identical fill and byte-identical questions, drains its attention mass and costs accuracy together. (b) Setting that mass directly, with the evidence left in place, reproduces the penalty and traces a smooth graded dose-response—no threshold. The diamond is the independently-run removal experiment’s endpoint (agreement 0.004 layer-24-indexed, 0.017 all-layer). (c) At distance 0 and fixed fill, near-duplicate accumulated context costs a further 0.085. Bootstrap CIs resample cases throughout.

prefix and suffix) instrument adherence in the accumulation and clamp arms. Grader rules and filler construction are in Appendix C.

Statistics. Accuracy is a fraction throughout, “points” are percentage points, and the joint-fit coefficients are in fractions per unit of their predictor (Appendix E). Per-condition n is bounded by the context window; every clamp experiment is paired over shared items and analysed with a paired bootstrap resampling cases (10,000 draws), never heads or turns. Resampling arms independently inflates intervals about $2.5\times$. An overflow guard skips rather than truncates any item that would not fit whole, since truncation at the window edge manufactures exactly the late-window cost under study (per-arm skip counts in Appendix E).

4 Results

4.1 Accumulation shows no net cost

Per-case accuracy does not fall as context fills. It is worst at the cold start and warms up ($\text{corr}(\text{correct}, \text{fill}) = +0.11$ Instruct, $+0.07$ DPO). In-context learning on a homogeneous task could mask a hidden decay. A random-subject stream removes this by drawing each question uniformly from all 57 MMLU subjects, leaving only the answer format learnable from prior context. Accuracy is still flat: its correlation with fill is -0.02 (random, $n=728$) and $+0.032$ (coherent, $n=988$), computed over fill < 0.8 . At 95% the data exclude declines larger than 2.9 points (coherent) and 5.5 (random). Canary instruction adherence stays at ceiling throughout the program’s one adherence arm (Qwen2.5-7B-Instruct, full 32k window). Adherence survives a compliant history. It does not survive a history that demonstrates a competing format (§4.4), and that contrast is the subject of §4.4–§4.5. Accumulation also drives the current query’s attention share down to ≈ 0.15 , the level at which clamping the same share on a cold-start context begins to cost accuracy (Appendix G). That accuracy stays flat anyway is consistent with in-context learning compensating for the lost attention. The accuracy null replicates on Qwen2.5-7B (Appendix J).

4.2 Displacement acts through attention mass

We now vary only the position of the answer-bearing evidence. Each DDXPlus probe’s vignette is placed k user turns before its question. Mean fill is 0.688 in every arm, the question text is byte-identical, and the overflow guard skipped 0 of 192 probes (example transcript in Appendix D). Every gap’s CI excludes zero (Table 9, Fig. 2a). In a joint fit of accuracy on fill and distance, distance carries the coefficient and fill carries none. Within the local arm, the one that reproduces the accumulation design, accuracy is flat with fill (Table 6). The effect survives restriction to parsed responses and is then fully monotone ($0.524 \rightarrow 0.413 \rightarrow 0.397 \rightarrow 0.343 \rightarrow 0.338$). The decline saturates by $k \approx 10$, where the $k=10$ and $k=20$ arms stop being distinguishable. The same model, on the same items at the same fill, degrades only when the evidence is displaced. These same items degrade as soon as dilution is reintroduced, which places §4.1’s flat accuracy away from any ceiling.

Table 1: **Removing the evidence’s attention mass reproduces the displacement penalty; restoring it does not.** Paired contrasts in percentage points, all-layer shares, bootstrap 95% CIs over shared items ($n=192$; the sweep at $n=131$). Positive is accuracy lost.

contrast	Δ accuracy (pts)	95% CI	of penalty
local – clamped to back_20 share	+15.1	[+9.9, +20.8]	91%
clamped – back_20	+1.6	[−5.2, +8.3]	—
natural – sweep endpoint at displaced share	+16.8	[+10.2, +23.4]	—
restored to local share – back_20	+4.7	[+1.0, +8.3]	28%
per-head restoration – uniform clamp	−2.1	[−8.3, +4.2]	—

Distance and attention drain move together by construction. Table 9 is consistent with dilution but does not establish it, because a purely positional account predicts the same table. Four interventions separate them.

Observational correlations. Evidence share falls $0.0408 \rightarrow 0.0124$ across the ladder ($r = -0.83$ with distance) while the question’s share stays flat, confirming only the evidence moved. One caution: within an arm, higher evidence share predicts lower accuracy ($\beta = -11.2$ per unit share, and stronger controlling for vignette length). Attention there tracks item difficulty (the model looks harder at confusing vignettes) and has the opposite sign to the causal effect below.

Mass removal. Keeping the evidence at local and clamping its share down to the same item’s own back_20 share reproduces the displacement penalty and lands the arm on back_20 (Table 1, paired $n=192$). **Mass removal alone reproduces the displacement penalty, with nothing moved.**

Dose-response. Sweeping the evidence’s share at fixed position over seven levels (layer-24-indexed, common subset $n=131$ present at every level) gives a graded decline from 0.473 at the natural share (0.0441) to 0.275 at 0.012 (Fig. 2b). Six of the seven levels differ significantly from the natural share, including a cut of less than a fifth of the mass, and no step is a threshold: the largest adjacent step is 0.053 and the decline is smooth throughout. Its endpoint at the displaced share matches the independently-run removal experiment in both denominations—to 0.4 points layer-24-indexed and 1.7 points on Table 1’s all-layer scale.

Mass restoration. The converse is only partial: clamping the displaced evidence’s share back up to its local level recovers little of the penalty (Table 1). The natural suspect is the instrument, since a uniform clamp cannot reconstruct which heads should carry the restored mass. A pattern-matched test rules that out: one closed-form bias per query head per layer, restoring each item to its donor’s per-head pattern (mean per-head share error 0.004, bias SD 1.22), gains nothing over the uniform clamp (Table 1). Head identity is not what restoration is missing. One bias per head is still uniform over the evidence’s ~ 280 tokens, and the remainder is either that within-span token pattern or a positional channel outside attention altogether. The drain itself is positional in tokens, not turns. A 64% mass cut from token distance alone costs at most 9.4 points, consistent with RoPE distance decay (Su et al., 2024; Wu et al., 2025) rather than conversation structure (Appendix G). The ladder, the sufficiency clamp (107% of the gap), and the graded dose-response all replicate on Qwen2.5-7B, there with no unparsed replies (Appendix J).

4.3 Competition acts through the competitors, not the evidence

Burying a needle varies two things at once: the evidence becomes distant and it becomes surrounded by plausible alternatives. We isolate the second by holding the evidence at the current query and fill fixed, and varying only how confusable the accumulated context is. Every arm accumulates eight prior DDXPlus cases, each answered in the transcript with its gold letter, so format and in-context-learning affordance are constant. The arms differ only in how much the accumulated cases’ differentials overlap the probe’s five options: random (the natural stream, 0.80 shared options on average), disjoint (zero), and near_dup (at least 3, averaging 3.75). No arm admits a context case with the probe’s own gold, so the manipulation is that in near_dup the probe’s gold routinely appears as some context case’s distractor (Appendix C). Paired over $n=365$ probes, accuracy is random 0.512, disjoint 0.485, near_dup 0.427 (Fig. 2c).

Table 2: **What the closure recovers depends on what the closed tokens say, not on how much attention they hold.** Generation-time closures on the near-duplicate arm, paired $n=365$, every arm closing the same ≈ 128 -token budget. Mass closed is a share of the final-position attention row; net accuracy is in percentage points against each arm’s size-matched random control. Bold marks an interval excluding zero.

closed set	mass closed	net accuracy (pts)	95% CI
verbatim option-name mentions	0.77%	+6.0	[+0.8, +11.2]
verbatim, re-run alongside the arms below	0.76%	+6.9	[+1.6, +12.1]
most-read content tokens	8.7%	+0.3	[−4.9, +5.5]
most-read tokens including template glue	41.6%	−17.5	[−23.0, −12.1]

199 Near-duplicate context costs 8.5 points against the natural stream and 5.8 against disjoint context
200 (intervals in Table 7). In a joint fit within this experiment, option overlap carries the coefficient
201 and fill carries none (Table 6), the same shape as the distance result on a different variable. Two
202 controls hold: the low-competition arms land on §4.2’s local accuracy of 0.464 (both gaps n.s.),
203 and disjoint is the longest arm while near_dup is the shortest; length works against the result.
204 The effect survives restriction to parsed responses (+8.2 points, significant). It is not monotone in
205 confusability, with random sitting above disjoint (n.s.): near-duplicate context is costly, and the
206 cost does not track overlap.

207 **The dissociation on mass.** Repeating the sweep with attention capture (same probes, same seeds)
208 separates the two mechanisms on mass: displacement removes 4.55% of the evidence’s attention share
209 and competition removes 0.22%, while costing 18.8 and 8.5 accuracy points respectively—a $\sim 20\times$
210 smaller mass change at half the accuracy cost, far too small to act through the displacement dose-
211 response. The two drain profiles are negatively correlated across layers ($r = -0.32$): displacement
212 bites hardest at layers 0–3 and 16, competition at 17–18—opposite attention signatures for the
213 same accuracy contrast. Per-head structure, including the null result that no head hides a large
214 countervailing move, is in Appendix F.

215 **The competitor-closure test.** An evidence-mass change this small rules out starvation as the carrier.
216 It does not touch the other attentional route, in which the filler’s (arm-constant) mass lands on
217 instances of the probe’s own answer candidates and reading them is what misleads. We test this
218 directly: closing every context occurrence of the probe’s option names at generation time recovers
219 **56%** of the competition penalty (Table 2), the residual gap to random no longer excludes zero, and
220 the rescue survives restriction to parsed responses. **Competition is attention-mediated through**
221 **the competitors, not the evidence:** less than 1% of the attention distribution carries the effect, and
222 removing it removes most of the cost. Both accuracy mechanisms are attentional, on different spans:
223 displacement removes mass from the evidence, competition directs mass onto the competitors, and
224 the evidence-mass measurement dissociates them because it is blind to the second by construction.
225 The $\sim 40\%$ residual is not instrument slack of the high-attention kind: closing the most-read content
226 tokens of the context body, the same number of tokens as the mentions and 8.7% of the row’s mass to
227 their 0.76%, recovers nothing (Table 2). The closure works through what the closed tokens say, not
228 the mass they receive, and the residual points at prefill-borne interference. The same measurement
229 concentrates 42% of the entire final-position row in the ≈ 128 most-read tokens, dominated by the
230 chat template’s turn delimiters. Closing that set is broadly destructive, but what it disrupts is the
231 demonstrated reply shape, not competition (§4.5).

232 **The family divergence.** Competition alone does not carry across families, and the divergence is
233 itself informative. On the same $n=365$ panel Qwen2.5-7B shows no detected penalty (random –
234 near_dup = +3.0 points), and the scale-0 competitor closure that recovers 56% of OLMo’s penalty
235 does nothing there (+1.9 points, random-closure control 0.000). The attention signatures separate
236 with disjoint intervals. Under near-duplicate context OLMo’s evidence share is flat (+0.0003, null)
237 while Qwen’s rises (+0.0071 in share, a disjoint interval), as if that model defends the evidence
238 under confusability rather than merely not paying for it (Fig. 8c,d). The families differ not by
239 degree but in whether the competitor-reading channel is engaged at all. We treat competition as
240 architecture-indexed and rest its positive claims on OLMo (Appendix J).

Table 3: **The instruction’s attention decides the contest only while the transcript demonstrates nothing applicable.** $n=40$ per cell. Accumulation alone drives the system span’s share $0.166 \rightarrow 0.021$ over eight turns, so row 2’s clamped level is one the stream reaches by itself.

condition	compliance	accuracy	what it shows
<i>no assistant turn in context</i>			
system span at its natural 0.165	0.99	0.525	the instruction governs a silent transcript
clamped to share 0.050	0.03	0.467	its attention is causally necessary
<i>accumulated filler, at matched fill ≈ 0.5</i>			
code, no applicable shape	1.00	—	erosion tracks how far the demonstrated
gsm8k, loosely applicable	0.60	—	shape transfers to the probe, not the
mmlu, a drop-in replacement	0.00	—	instruction’s attention
<i>at depth 42, reversing it</i>			
natural	0.000	0.650	the eroded floor
clamp the system span back up	1.000	0.425	re-weighting reverses it, zero-sum in accuracy
restate the instruction	1.000	0.450	re-presenting reverses it, without surgery
both levers	1.000	0.275	the two costs add

241 4.4 Precedent: in-context learning against the system prompt

242 The costs so far are to accuracy. A third mechanism costs compliance: not whether the model can
 243 still solve the task, but whether it answers in the instructed form (§3, $n=40$ probes per depth, one
 244 accumulation snapshot at every depth, so depth is the only mover).

245 Every stream here is also a stream of in-context examples. Each prior case is answered in the
 246 transcript: the history demonstrates an answer shape while the system prompt instructs one, two
 247 channels feeding one decision. When they agree, as in §4.1’s compliant history, nothing surfaces.
 248 This section makes them disagree, and the instruction gives way (cf. Anil et al., 2024).

249 **Attention decides the contest only when nothing is demonstrated.** In a context with no assistant
 250 turn, clamping the system span collapses compliance with accuracy unchanged and parsing at 0.967
 251 (a suffix canary flips on 120 of 120 items). The clamped level is one accumulation passes on its own
 252 between two and four turns (-2.6 nats), well clear of the near-ablation regime excluded in §4.2. This
 253 confirms Dongre et al. (2026)’s ablation finding that system-prompt attention is causally necessary,
 254 and sharpens it: the collapse needs no masking, only a graded clamp to a share accumulation reaches
 255 by itself. Put one example in the transcript and the clamp stops mattering. A single compliant example
 256 pins compliance at ceiling at every clamp level and a single bare-letter example pins it at floor even
 257 unclamped, replies byte-copying the previous assistant turn’s format at zero length variance across
 258 720 generations per arm (Appendix H). The instruction governs behaviour only when the transcript is
 259 silent.

260 **What decides it instead is applicability.** Three filler streams vary how far the demonstrated
 261 shape transfers to a 5-option probe: code (request $\rightarrow$ Python function), gsm8k (worked prose), mmlu
 262 (MCQ $\rightarrow$ bare letter). Compliance follows that ordering rather than attention. mmlu collapses $0.875 \rightarrow$
 263 0.000 within three filler turns (fill 0.147) and never recovers through fill 0.87, while code still holds
 264 0.900 at fill 0.778 (Fig. 4a). Accuracy is unharmed throughout: the model still solves the case and
 265 answers in the demonstrated style. The system prompt’s attention does not carry the difference, since
 266 its raw share falls on the same arithmetic curve in every arm, its per-token enrichment is flat-to-rising
 267 in all three (Fig. 4b), and code holds full compliance at half the raw share at which mmlu has already
 268 collapsed. The same model at the same near-zero share survives or collapses depending on what the
 269 filler demonstrates, an axis Dongre et al. (2026)’s fixed filler cannot see. Natural dilution does not
 270 cause the erosion, being identical in the arm that never erodes.

271 Either lever reverses it in full (Fig. 4c), which makes attention mass a causal weight in the contest:
 272 clamp the instruction down and it loses unopposed, clamp it up and it wins against 42 counter-
 273 exemplars, leave it natural and the demonstrated template beats it. Re-presenting beats re-weighting,
 274 since mass forced onto a 48-token span is taken from everything else by §4.2’s zero-sum arithmetic
 275 while a restatement costs the model nothing. The applicability ordering, the reading signature, and
 276 restatement’s advantage all replicate on Qwen2.5-7B, where the erosion is staged rather than total
 277 (Appendix J).

Table 4: **The mode is readable and installable, but not reachable through attention to the demonstrated answers.** Compliance at depth 42 ($n=40$ per arm) unless noted; closures are scale-0 at generation time against size-matched random controls. Enrichment is `mm1u/gsm8k/code`, the erosion ordering of §4.4.

operation	outcome	what it shows
<i>on attention to the demonstrated answers</i>		
read out, per-token enrichment	+1.28/+0.34/neg.	they are read in the erosion order
close at decode	compliance 0.000	reading does not sustain the mode
close at prefill, decode, or both	0.000 in every window	reading does not install it
the same closure, on accuracy	−17.5 pts prefill, 0.0 decode	the reading is worth accuracy, not format
<i>on the residual stream</i>		
probe the instruction’s presence	transfer AUC 1.000	represented at every depth, overruled
probe the upcoming reply	LOO AUC 0.875	the outcome is fixed before the first token
steer, mean-difference vector	0.000 vs. 0.425 control	the mode is installable by direction
erase, four projections	every one null	it is not erasable as a direction
patch the template glue positions	62% of the full patch	the carrier is positional

4.5 The mode: readable, installable, not erasable

We call the behavioral state §4.4 leaves behind the mode: the disposition to answer in the demonstrated shape rather than the instructed one. Table 4 is the search for it. Nine operations divide cleanly. Every operation on attention to the demonstrated answers leaves compliance at the eroded floor, in any window. Every operation on the residual stream moves it, reads it, or fails only at erasure.

That split is the sharpest dissociation in the paper, because the operation on the left of it is the one that worked in §4.3: the same scale-0 closure that recovers 56% of the competition penalty restores nothing here, in any window (Fig. 5). Competition is generation-time reading. The precedent mode is in place by prefill, a template already installed, task-vector style (Hendel et al., 2023), through some channel other than attention to the answer spans. The probe rows say the instruction is read, represented, and overruled, and that the outcome is decidable from the residual stream before the first token is emitted. The erase nulls are not redundancy: two fitted projections drive the layer’s linear code to chance while the behavior is untouched (Appendix H), so the code is low-rank and the behavioral decision does not consume the linearly decodable component. We can decode the mode and we can install it. Neither shows what carries it.

The carrier, localized in tokens. What direction-level search could not find, position-level search does—on the second family, where a counterfactual patching instrument exists (Appendix J): transplanting per-position hidden states between two transcripts identical except for their format instruction, and reading out the instructed formats’ log-probability contrast (Table 11). The chat template’s turn delimiters (the `im_end/im_start` glue, holding no instruction text and no demonstrated answer) carry 62% of the full-context patch, every content subset carries at most 12%, and the crossed cell of glue positions at layers 7–13 still carries 42%. The format state rides the transcript’s skeleton, tokens the chat template already supplies, consistent with delimiter and low-information tokens serving as computation slots (Appendix A), and it is where the OLMo attention row concentrates (§4.3, Fig. 3). Unlike the mass absorbed by attention sinks (Xiao et al., 2024; Gu et al., 2025), which is content-independent normalization slack, the glue state is content-bearing and transportable: moving it moves the behavior. The channel is cross-family. On OLMo, whose full-transcript effect is an order of magnitude smaller and not inverted, the glue-only patch carries all of it (+0.092, $n=85$ after skips, vs. the full patch’s +0.056, $n=34$, with the unrelated-fact control at −0.011 in both).

4.6 Why context rot gets reported: the signatures

A generic depth effect would survive on heterogeneous dialogue, and the entropy collapse behind the context-rot reports does not. On 150 organic WildChat conversations (Zhao et al., 2024), within-conversation entropy is flat with depth (median late/early ratio 0.99). Partitioning a second, 400-conversation run by its own inter-turn homogeneity separates the driver: homogeneous tertiles collapse (0.897 late/early) while heterogeneous ones are flat (1.001, $n=134$ each). Because homogeneous conversations are longer, length works against the effect. Across OLMo-2’s (Team

OLMo, 2024) post-training chain the collapse is mostly a downward level shift in baseline entropy installed by alignment, not a steeper slope (Appendix I).

The residual hazard in the localized case is calibration, not competence. On the DDXPlus streams logging per-token confidence ($n=154$ pooled cases, two models, both question orders), answer confidence rises $0.85 \rightarrow 0.95$ with fill ($r = +0.72$) at flat accuracy, and equally on wrong answers ($0.86 \rightarrow 0.95$, $r = +0.69$): confidence in wrong answers rises ~ 10 points over a session (Fig. 6c), most consequential where a homogeneous high-stakes stream makes the model surest.

5 Conclusion

With accumulation, displacement, competition and precedent separated inside one harness, each mechanism leaves a distinct signature:

- **Accumulation** produces a monotone reallocation of attention and a collapse of predictive entropy, but no detected loss of accuracy and no loss of instruction adherence—the with-length decline that motivates the rot framing (Liu et al., 2023; Hong et al., 2025; Laban et al., 2025) does not surface in these streams.
- **Displacement**, the retrieval case, produces a large accuracy cost that is reproduced in full by removing the evidence’s attention mass, graded in that mass, and positional in tokens. This is the one mechanism that is attention dilution proper.
- **Competition** produces a further cost while the evidence’s attention barely moves: interference that raw context length does not index (cf. Wang and Sun, 2025), carried instead by attention to the competing instances themselves, under 1% of the distribution.
- **Precedent** erodes not accuracy but compliance, in a contest the instruction loses with its attention intact: the benign, self-generated counterpart of demonstrations overriding priors and instructions (Wei et al., 2023; Anil et al., 2024; Camassa and Shiller, 2026).

The literature’s central claim is correct but misattributed and undercounted. In these streams, context rot is not a decay with length: it is three mechanisms with different signatures, of which length is merely the usual way of causing all three at once. A needle buried far away among plausible alternatives moves the displacement and competition levers at once, and length-indexed benchmarks cannot separate them (Hsieh et al., 2024b); attention-share accounts of multi-turn decline (Dongre et al., 2026) describe only the mechanism that is dilution proper. The precedent state itself, unreachable as a direction, is reachable as a place: the transcript’s skeleton electing itself as native instruction storage (Darcet et al., 2024; Sun et al., 2024; Mu et al., 2023). Displacement, its mass sufficiency, the accumulation null, and the precedent contest replicate on a second model family. Competition is the mechanism that does not: it is architecture-indexed, its attention signature inverting on Qwen (Appendix J).

6 Limitations

Per-condition n is bounded by the window, and every interval depends on §3’s paired analysis. The accumulation null resolves declines above 2.9 points (coherent) and 5.5 (random); its random-subject control leaves format and protocol learning available in both streams. For displacement, removal establishes sufficiency in full while restoration returns 0.28 of the penalty. Head-uniformity is excluded as the clamp’s limitation, but it still spreads restored mass uniformly over the span’s tokens, leaving a within-span token pattern and a non-attention positional channel unseparated. The competition result is one significant step rather than a gradient: mild overlap is indistinguishable from none, and the near-duplicate arm changes the selection regime. Its closure rescue leaves $\sim 40\%$ of the gap unexplained, and the hot-set control, ranked on the all-layer mean row, leaves a low-mass paraphrase channel open. The instruction-canary null is a floor effect, the WildChat partition is observational with no uncertainty on the tertile contrast, competition rests on one family, and the low-share compliance dissociation is OLMo-specific (Appendix J).

The precedent results use one instruction family at $n=40$ per cell, and the filler streams co-vary in genre, structure, and reply length, leaving the applicability ordering resting on the stream contrast with the one-example and reversal experiments. Window-split closure rules out exemplar-answer attention for that span family only. The carrier localization is confounded with positional periodicity, its patch readout is teacher-forced, and the mechanisms are ranked at §3’s 4k window rather than at 128k.

References

- R. Agarwal et al. Many-shot in-context learning. *NeurIPS*, 2024.
- G. Alain and Y. Bengio. Understanding intermediate layers using linear classifier probes. *arXiv:1610.01644*, 2016.
- C. Anil et al. Many-shot jailbreaking. *NeurIPS*, 2024.
- Y. Belinkov. Probing classifiers: promises, shortcomings, and advances. *Computational Linguistics*, 48(1), 2022.
- A. Camassa and B. Shiller. Do as I say, not as I do: instruction-induction conflict in LLMs. *arXiv:2605.20382*, 2026.
- F. Cuconasu et al. The power of noise: redefining retrieval for RAG systems. *SIGIR*, 2024.
- T. Darcet, M. Oquab, J. Mairal, and P. Bojanowski. Vision transformers need registers. *ICLR*, 2024.
- G. Davidson, T. Gureckis, B. Lake, and A. Williams. Do different prompting methods yield a common task representation in language models? *NeurIPS*, 2025.
- V. Dongre et al. When attention closes: how LLMs lose the thread in multi-turn interaction. *arXiv:2605.12922*, 2026.
- A. Grattafiori et al. The Llama 3 herd of models. *arXiv:2407.21783*, 2024.
- X. Gu et al. When attention sink emerges in language models: an empirical view. *ICLR*, 2025.
- R. Hendel, M. Geva, and A. Globerson. In-context learning creates task vectors. *Findings of EMNLP*, 2023.
- D. Hendrycks et al. Measuring massive multitask language understanding. *ICLR*, 2021.
- K. Hong, A. Troynikov, and J. Huber. Context rot: how increasing input tokens impacts LLM performance. Chroma technical report, 2025.
- C.-Y. Hsieh et al. Found in the middle: calibrating positional attention bias improves long context utilization. *Findings of ACL*, 2024.
- C.-P. Hsieh et al. RULER: what’s the real context size of your long-context language models? *COLM*, 2024.
- P. Laban, H. Hayashi, Y. Zhou, and J. Neville. LLMs get lost in multi-turn conversation. *arXiv:2505.06120*, 2025.
- M. Levy, A. Jacoby, and Y. Goldberg. Same task, more tokens: the impact of input length on the reasoning performance of large language models. *ACL*, 2024.
- N. F. Liu et al. Lost in the middle: how language models use long contexts. *TACL*, 2023.
- T. Lieberum et al. Gemma Scope: open sparse autoencoders everywhere all at once on Gemma 2. 2024.
- K. Meng, D. Bau, A. Andonian, and Y. Belinkov. Locating and editing factual associations in GPT. *NeurIPS*, 2022.
- J. Mu, X. L. Li, and N. Goodman. Learning to compress prompts with gist tokens. *NeurIPS*, 2023.
- N. Mu et al. A closer look at system prompt robustness. *arXiv:2502.12197*, 2025.
- C. Olsson et al. In-context learning and induction heads. *arXiv:2209.11895*, 2022.
- F. Shi et al. Large language models can be easily distracted by irrelevant context. *ICML*, 2023.
- J. Su et al. RoFormer: enhanced transformer with rotary position embedding. *Neurocomputing*, 568, 2024.
- M. Sun, X. Chen, J. Z. Kolter, and Z. Liu. Massive activations in large language models. *arXiv:2402.17762*, 2024.
- A. Fansi Tchango et al. DDXPlus: a new dataset for automatic medical diagnosis. *NeurIPS Datasets and Benchmarks*, 2022.
- Team OLMo et al. 2 OLMo 2 Furious. *arXiv:2501.00656*, 2024.
- E. Todd et al. Function vectors in large language models. *ICLR*, 2024.

407 J. Vig et al. Investigating gender bias in language models using causal mediation analysis. *NeurIPS*, 2020.

408 C. Wang and Y. Sun. Unable to forget: proactive interference reveals working memory limits in LLMs beyond
409 context length. *ICML Workshop on Long-Context Foundation Models*, 2025.

410 K. Wang, A. Variengien, A. Conmy, B. Shlegeris, and J. Steinhardt. Interpretability in the wild: a circuit for
411 indirect object identification in GPT-2 small. *ICLR*, 2023.

412 J. Wei et al. Larger language models do in-context learning differently. *arXiv:2303.03846*, 2023.

413 X. Wu, Y. Wang, S. Jegelka, and A. Jadbabaie. On the emergence of position bias in transformers. *ICML*, 2025.

414 G. Xiao et al. Efficient streaming language models with attention sinks. *ICLR*, 2024.

415 A. Yang et al. Qwen2.5 technical report. *arXiv:2412.15115*, 2024.

416 F. Zhang and N. Nanda. Towards best practices of activation patching in language models: metrics and methods.
417 *ICLR*, 2024.

418 Q. Zhang et al. Tell your model where to attend: post-hoc attention steering for LLMs. *ICLR*, 2024.

419 W. Zhao et al. WildChat: 1M ChatGPT interaction logs in the wild. *ICLR*, 2024.

420 **Code and artifacts.** The harness, the span-attention clamp, every driver behind the experiments,
421 and the committed run artifacts (per-turn CSVs and summaries) that each number in this paper traces
422 to are at <https://anonymous.4open.science/r/Breaking-Down-Context-Rot-A553/>.

423 A Background and related work

424 **Length as the suspected cause.** Most evidence for context rot comes from setups where the
425 answer-bearing information sits inside a single long input. Liu et al. (2023) showed that accuracy
426 depends strongly on where in the context the relevant passage sits, dipping when it falls in the middle.
427 Levy et al. (2024) held a reasoning task fixed and padded the input with task-irrelevant text, finding
428 degradation well before the models’ advertised context limits. Hong et al. (2025) ran a broad sweep
429 over 18 models and found that reliability falls with input length even on tasks as simple as retrieval
430 and replication, and that the fall is not uniform: it depends on how similar the needle is to the
431 question, and on whether distractors are present. Hsieh et al. (2024b) show the benchmark-level
432 version: near-perfect needle scores coexist with sharp degradation on harder long-context tasks, and
433 claimed context sizes overstate usable ones. Laban et al. (2025) report a large multi-turn degradation,
434 but locate its cause in the model committing early to a wrong reading and failing to recover—not in
435 length as such.

436 **Confusable context as a separate suspect.** A second literature manipulates the content of the
437 surrounding context rather than its length. Shi et al. (2023) insert an irrelevant sentence into
438 grade-school math problems and observe a sharp accuracy drop. In retrieval-augmented generation,
439 Cuconasu et al. (2024) find that documents retrieved as relevant but not containing the answer act
440 as distractors and degrade accuracy. Wang and Sun (2025) show accuracy declining log-linearly as
441 interfering similar updates accumulate, independent of raw context length—the nearest neighbor to
442 our competition result, with no attention instrumentation. Our contribution is to locate the cost: it
443 runs at held-constant evidence attention mass and is carried by attention to the competing instances
444 themselves (§4.3).

445 **Instructions versus demonstrations.** Demonstrations can override a model’s trained priors (Wei et
446 al., 2023) and, at scale, its safety training (Anil et al., 2024). The same lever is framed as pure benefit
447 in many-shot prompting (Agarwal et al., 2024). Hardcoded assistant turns exhibiting a competing
448 pattern flip instruction-following into pattern-following (Camassa and Shiller, 2026), system-prompt
449 adherence fails under conflicting or adversarial user turns (Mu et al., 2025), and instruction-derived
450 and demonstration-derived task representations are measurably distinct (Davidson et al., 2025). Our
451 eroding pressure differs from all of these: it is benign, self-generated, and contains no conflicting
452 directive anywhere. It is precedent, not conflict or attack, and we identify the contest’s causal weight
453 and both of its reversals.

Attention as measurement and lever. Closest to our mechanistic question, Dongre et al. (2026) track attention from responses to the system prompt declining over 50 filler turns across four model families, show that hard-masking the instruction out of the window collapses compliance, and decode goal information from the residual stream at high AUC after attention has “closed”. Our design differs in three ways. Their goal-accessibility metric aggregates raw share, which falls by arithmetic as context grows (§2.2). Our token-normalized enrichment removes that decline entirely, and the corrected quantity is flat in eroding and non-eroding arms alike. Their causal arm is total ablation (attention necessary, which our system-span clamp confirms), while our clamp is graded and bidirectional, and the restoration direction (compliance recovered against 42 counter-exemplars) is an arm ablation cannot supply. Their filler is always inapplicable. Our applicability ladder shows the same model at the same near-zero share surviving or collapsing depending on what the filler demonstrates, a content dependence invisible to fixed-filler designs. On the steering side, Zhang et al. (2024) boost attention on profiled head subsets to improve compliance and Hsieh et al. (2024a) recalibrate a static positional attention bias globally. Our clamp is the measurement-grade counterpart (bit-identical no-op, solvable dose, span-targeted in accumulating multi-turn context), used to establish when attention mass is and is not the mediating variable. Position-driven attention structure is the content-independent backdrop to all of this: sink tokens absorb mass regardless of content (Xiao et al., 2024; Gu et al., 2025), and the causal mask plus RoPE produce distance decay by construction (Wu et al., 2025). Our clamps and token-normalized shares target content spans against that backdrop.

Compact task representations. In-context demonstrations compress into residual task vectors (Hendel et al., 2023) transported by identifiable heads (Todd et al., 2024). These are the no-attention-signature channel our precedent mechanism appears to use (§4.5), and our erase-null results bound what such compact readable structure can be assumed to do causally. The byte-exact format copying our pinned arms show (Appendix H) is the behavior induction heads implement at circuit level (Olsson et al., 2022). We implicate that circuit behaviorally and at span level, without head-level verification.

B Instrument validation

Attention capture. Shares are read from a reconstruction of each layer’s attention at the final position, which avoids materializing $N \times N$ attention matrices. With $\tilde{q}$ and $\tilde{K}$ the final-position query and the keys after QK-norm (where the architecture applies it) and RoPE, and m the additive attention mask, the reconstructed row is

$$a = \text{softmax}\left(\frac{\tilde{q} \tilde{K}^\top}{\sqrt{d}} + m\right).$$

The reconstruction matches the framework’s own output_attentions rows to $\max |\Delta| = 1.1 \times 10^{-6}$ on OLMo-2 and 3.1×10^{-5} on a GQA architecture (Qwen-2), both in fp32. Including m makes the capture valid under a mask-based intervention: a capture that ignores the mask still matches on plain forwards, where a causal mask’s last query row is all zeros, while silently reporting unclamped attention during a clamp.

Clamp. At scale 1.0 the clamp is bit-identical to the unhooked model ($\max |\Delta| \text{ logits} = 0.0$), because it adds $b=0$ to the existing mask rather than reimplementing attention (§2.3). The bias is uniform across heads, and a span’s aggregate share is hit by bisection on b . All solved levels in this paper land within $\sim 10^{-4}$ of target. Setting a span’s scale to 0 fills its key columns with the mask minimum—closure, the same operation as a causal mask. Interpretation is restricted to biases well clear of near-ablation: levels requiring -4.7 to -6.1 nats produce degenerate behavior (the model answers “A” on 103–110 of 110 items) and are excluded.

C Task and filler specifications

Probe construction. DDXPlus probes are 5-option differentials with the vignette as evidence. Options are shuffled per probe: DDXPlus lists the differential in rank order, so unshuffled gold is the letter “A” in 71.4% of probes, and arms that differ in letter bias then differ in accuracy for free. Items with fewer than 5 options are excluded (26.7% of raw cases, some answerable from a single

option without the vignette). Generation budgets are per-protocol: the accuracy arms (§4.2–§4.3) generate 32 new tokens. The compliance arms (§4.4) generate 128 for the probe reply (fillers up to 200), because shorter caps truncate the reply component that formats put last, biasing exactly the metric under study. The probe and steering runs (Appendix H) generate 256.

Format instruction and grader. The clinical template requires ANSWER: with a letter and SUPPORTING: with at least two findings quoted from the vignette. Compliance requires the template. Accuracy is scored separately and more permissively (a letter offered in prose, or alone on the reply’s first line, is scoreable for accuracy while remaining non-compliant). The grader was validated against real generations from every condition. Graders written against imagined output formats systematically mis-score exactly the conditions where the treatment changes the output format.

Filler streams. Filler replies are the model’s own (up to 200 tokens, mmlu answers truncated to the letter), so each arm demonstrates the shape the model itself produces for that stream rather than a shape we wrote. Table 5 gives the three. Applicability is the design variable: it is the only thing that separates the arms, and it is what §4.4 shows the erosion follows.

Table 5: **The three filler streams, ordered by how far their demonstrated answer shape transfers to a 5-option probe.** Fill per turn differs by $\approx 3\times$, so the arms are compared at matched fill rather than matched depth.

stream	source	demonstrated reply	applicable to the probe?	fill/turn
code	30 synthetic request→function	a Python function	no shape in common	~ 0.056
gsm8k	grade-school math	worked prose	loosely, via prose	~ 0.030
mmlu	4-option MCQ, non-medical	a bare letter	a drop-in replacement	~ 0.018

Canaries. The system-clamp arms use a three-part canary instruction (reply prefix marker, suffix tag, and a forbidden phrase). The adherence arm uses the prefix and suffix pair. The forbidden-phrase canary is satisfied trivially by short replies, so two canaries are informative.

Run configuration. The settings that hold across the whole program, gathered here because they are otherwise stated where they are first used. The primary model is allenai/OLMo-2-1124-7B-Instruct and the replication model Qwen/Qwen2.5-7B-Instruct, both at seed 42 with greedy decoding. Every mechanism experiment runs in a 4k window on both families; the two departures are not mechanism results, the adherence arm of §4.1 at 32k and the WildChat measurements at 16k and 32k. An attention share is the post-softmax mass at the final position, meaned over heads and pooled over all 32 layers, with layer 24 read as a reference in the ladder diagnostics, and the clamp is uniform across heads and applied at every layer unless stated otherwise. All contrasts are paired over shared items, all intervals are case-resampled bootstraps at 10,000 draws (4,000 for fit coefficients), and the overflow guard skips any item that would not fit whole rather than truncating it. Probe construction and the per-protocol generation budgets are above; the drop counts are in Appendix E.

D Example transcripts and generations

Both examples are the first probe of their committed runs, reconstructed byte-exactly from the run’s seed and validated against the stored artifacts (token counts, pathology, gold). The generations are quoted verbatim from each run’s stored turns.csv, truncated where the 32-token generation cap truncated them. Transcripts are abridged, and [...] marks elided text.

Displacement (§4.2). Session 0’s first depth-21 probe: gold D (acute pulmonary edema), fill 0.50 in every arm. The transcript accumulates 21 answered MMLU filler exchanges. The vignette and the question are byte-identical across arms and only the vignette’s position differs—in local it is the user turn directly before the question, in back_10 the same turn sits ten filler exchanges earlier.

[user] You are answering a mixed stream of questions. Some are multiple-choice knowledge questions; some describe a patient. Answer each one as it arrives.

[assistant] Understood.

543 [user] Which of the following ions CANNOT be used as a dopant in germanium to make
544 an n-type semiconductor? A) As B) P C) Sb D) B. Reply with only the letter. . .

545 [assistant] D

546 [. . . 19 more filler exchanges . . .]

547 [user] Patient: 76-year-old Female. Chief complaint: Are your symptoms worse when
548 lying down and alleviated while sitting up. Symptoms: pain (tedious, sensitive, heavy) in
549 the lower and posterior chest, intensity 9, radiating to biceps, chin, jaw, shoulders; shortness
550 of breath; bouts of choking that wake you up at night; swelling in ankles, calves, tibia [. . .]
551 History: fluid in the lungs; dialysis; high blood pressure; heart attack or angina; heart failure
552 [. . .]

553 [assistant] Noted.

554 [user] For the patient described earlier, most likely diagnosis: A) Pulmonary embolism
555 B) Possible NSTEMI / STEMI C) Anaphylaxis D) Acute pulmonary edema E) Pericarditis.
556 Answer:

557 In back_10 the vignette turn and its Noted . sit above ten of the filler exchanges instead of none.
558 The generations for this item:

559 local: "D" (parsed, correct)

560 back_10: "Given the symptoms described for the patient, such as sudden onset of severe
561 chest pain that is worse when lying down and alleviated by sitting up, shortness of" (unparsed,
562 wrong)

563 The displaced reply paraphrases the vignette accurately, so the model is not blind to the evidence, but
564 it never commits to a letter. back_2, back_5 and back_20 fail the same way on this item. This is the
565 unparsed failure mode whose rate rises with fill. The ladder survives restriction to parsed responses
566 (§4.2), where the failures are wrong letters instead.

567 **Competition (§4.3).** Probe 0: gold E (viral pharyngitis). Every arm holds the vignette at the current
568 query (distance 0) and accumulates eight answered DDXPlus cases. Only option overlap differs. A
569 representative near_dup context case—four of its five options are the probe's, its gold is not:

570 [user] Patient: 64-year-old Male [. . .] Symptoms: [. . .] diffuse muscle pain; loss of
571 appetite; nasal congestion or a clear runny nose; a cough. History: smokes cigarettes;
572 is immunosuppressed [. . .] Most likely diagnosis: A) Influenza B) Bronchitis C) Viral
573 pharyngitis D) HIV (initial infection) E) URTI. Answer:

574 [assistant] A

575 [. . . 7 more such cases . . .]

576 [user] Patient: 13-year-old Male. Chief complaint: nasal congestion or a clear runny nose.
577 Symptoms: burning pain in both tonsils, thyroid cartilage, pharynx, under the jaw, intensity
578 4; fever; nasal congestion; a cough. History: contact with a person with similar symptoms;
579 lives with 4 or more people; is immunosuppressed [. . .]

580 [assistant] Noted.

581 [user] For the patient described earlier, most likely diagnosis: A) Bronchitis B) Acute
582 laryngitis C) Influenza D) URTI E) Viral pharyngitis. Answer:

583 The generations for this item:

584 disjoint: "E" (correct)

585 random: "E" (correct)

586 near_dup: "D The patient's symptoms include a cough, fever, and nasal congestion,
587 which are common in upper respiratory tract infections (URTIs). The fact that the" (parsed,
588 wrong)

589 Under near-duplicate context the model picks URTI, an option that recurs in six of the eight context
590 cases, over the gold, and justifies it from symptoms the context's cases share. With the same vignette
591 over disjoint or random context it answers correctly.

E Statistical procedures

All intervals are case-resampled bootstrap 95% CIs at 10,000 draws (joint-fit and slope coefficient intervals at 4,000). Clamp and arm contrasts are paired over shared items. The paired analysis cuts interval half-widths $\sim 2.5\times$ against independent resampling on the same data. Joint fits are linear probability models (accuracy in percentage points is the quoted scale) with case-resampled coefficient intervals. Probe analyses use leave-one-out cross-validation at episode level with 200-shuffle permutation nulls. The overflow guard skips items that would not fit whole (prompt + generation headroom) and reports per-arm counts: 0/192 in the distance sweep, 4 skipped +15 starved in competition (and the identical panel in the competitor-closure run), 0 in the mmlu erosion arm, and 7 in the code arm, all at its deepest depth; that ladder is interpreted to depth 12 (fill 0.778).

Table 6: **In both causal experiments the manipulated variable carries the coefficient and fill carries none.** Linear probability models on accuracy as a fraction, case-resampled bootstrap 95% CIs (4,000 draws). Bold marks an interval excluding zero.

fit	predictor	β	95% CI
displacement (§4.2)	distance, per turn	-0.0076	[-0.0117, -0.0035]
displacement (§4.2)	fill, per unit	-0.007	[-0.210, +0.187]
displacement, local arm only	fill, per unit	-0.294	[-0.767, +0.184]
competition (§4.3)	overlap, per shared option	-0.021	[-0.039, -0.003]

Table 7: Every interval quoted in the main text, in one place. Accuracy contrasts are in percentage points; the share contrast is in raw attention share. Paired case-resampled bootstrap 95% CIs. The joint-fit coefficients are in Table 6 and the mass interventions in Table 1.

quantity	family	value	95% CI
competition penalty, random - near_dup	OLMo	+8.5	[+3.0, +14.0]
competition penalty, disjoint - near_dup	OLMo	+5.8	[+0.6, +11.2]
exemplar closure during prefill, accuracy cost	OLMo	+17.5	[+7.5, +30.0]
competition penalty, random - near_dup	Qwen	+3.0	[-1.6, +7.4]
competitor closure rescue	Qwen	+1.9	[-2.5, +6.3]
evidence share under near_dup - random	Qwen	+0.0071	[+0.0062, +0.0080]

F Per-layer and per-head structure

Attention share is a mean over a layer’s 32 heads, and a mean can hold still while heads cancel underneath it. We measured all 1,024 heads under both accuracy-costing mechanisms to find out whether either aggregate hides structure. Table 8 is the answer: displacement is uniform enough that the mean is honest, and competition’s near-zero net is a real reallocation rather than inaction.

Table 8: **What the head-level measurement adds to each layer-24 mean.** Bonferroni-corrected per-head tests; the competition null is a paired sign-flip over 2,000 permutations.

	displacement	competition
layer-24 mean	-0.0284 evidence share	-0.0003, indistinguishable from zero
heads that move	32 of 32, all significant	19 of 32 significant
typical per-head move	fractional drain 0.689 (sd 0.147)	0.0026 against a null of 0.0004
depends on how much a head held?	no, $r=0.08$	—
reading	every head loses mass; the mean hides nothing	a reallocation of $\approx 6\%$ of the evidence’s local share, not inaction

One consequence for §4.2. A single uniform odds-scale reproduces 58% of the observed per-head displacement pattern, so the clamp’s head-uniformity is a closer approximation to the real drain than it might look. That weakens, without eliminating, the reading of the partial necessity result as pure instrument limitation.

Layer 24 is also not where evidence-specialised heads live. Such heads exist (255 of 1,024 attend the evidence more than its token share warrants, up to 0.626 for one head on a span occupying 0.102

of the context), but none of them sit at layer 24, whose mean evidence share (0.0408) is among the lowest in the model against 0.183 at layer 3 and 0.143 at layer 16. Head specialisation varies by layer, and one layer does not characterise it. Our layer-24 readouts are therefore diagnostics on a deliberately unexceptional layer, not a claim about where the model does its retrieval.

G Additional displacement analyses

Table 9: **Distance is the variable that matters** (OLMo-2-Instruct, $n=192$ per arm, chance = 0.200, fill 0.688 in every arm), the numbers behind Fig. 2a. Evidence attention share is measured at layer 24 on the same forwards. Case-resampled bootstrap 95% CIs.

arm	local	back_2	back_5	back_10	back_20
accuracy	0.464	0.359	0.292	0.250	0.276
cost vs local	—	+0.104	+0.172	+0.214	+0.188
95% CI	—	[+.005, +.203]	[+.073, +.266]	[+.120, +.307]	[+.094, +.281]
evidence share @L24	0.0408	0.0363	0.0320	0.0195	0.0124

Tokens versus turns. A partial 2×2 in (gap turns) $\times$ (filler length), total context equalized by padding, dissociates them: two arms matched on tokens have effectively identical evidence share (0.0104 vs 0.0108) despite a $4 \times$ difference in turn count, while at matched turns share falls $0.0288 \rightarrow 0.0104$ as the gap grows $446 \rightarrow 1684$ tokens—a 64% cut in the evidence’s attention that costs at most 9.4 accuracy points ($+2.6 [-4.2, +9.4]$). Share regresses significantly on gap tokens and not on gap turns. Attention drain is a function of token distance, as RoPE decay predicts. Conversation structure is not what drains it.

The current query’s floor. Accumulation drives the current query’s share from ≈ 0.35 down to ≈ 0.15 , and it is natural to ask whether that stops short of a harmful level. It does not: clamping the query span on cold-start contexts, accuracy is flat from the natural share (0.258, accuracy 0.545) down through 0.20 (0.509), then costs 16.4 points at 0.15 ($[+3.6, +29.1]$)—exactly the share accumulation reaches. Accumulation stops exactly at the edge of the flat region, not short of it. (Levels below 0.15 require -4.7 to -6.1 nats, near-ablation rather than points on a dose-response, and are excluded from interpretation.) The ladder re-denominates fractionally onto the all-32-layer mean share: natural 0.463, no cost above natural, flat at $0.775 \times$ natural ($+4.5$ points $[-1.8, +11.8]$), significant cost at $0.581 \times$ ($+11.8 [+2.7, +20.9]$), the same structure as at layer 24. In those units accumulation lands at 0.202 over eight cases, past the first costly level rather than at its edge. That accuracy stays flat under accumulation, while the same share reached by clamping costs points, is evidence that in-context learning is actively holding accuracy up.

Final-position attention, token by token (all-layer mean, log-scaled white→orange; solid underline = evidence, dashed = current question)

local – evidence share 0.0883

A) 1%
B) 3%
C) 6%
D) 10%

Reply with only the letter (A, B, C, or D).

<[assistant] |>
<[endofcontext]>
<[user] |>
Patient : 76-year -old Female
Chief complaint : Are your symptoms worse when lying down and alleviated while sitting up
Symptoms :
- Yes – Has pain somewhere , related to your reason for consulting
- Characterize your pain : tedious , sensitive , heavy
- Do you feel pain somewhere : lower chest , side of the chest (R), side of the chest (L), posterior chest wall (R), posterior chest wall (L)
- How intense is the pain : 9
- Does the pain radiate to another location : biceps (R), chin , jaw , shoulder (R), shoulder (L)
- How precisely is the pain located : 4
- How fast did the pain appear : 1
- Yes – Is experiencing short ness of breath or difficulty breathing in a significant way
- Yes – Has bouts of choking or short ness of breath that wake you up at night
- Yes – Has swelling in one or more areas of your body
- Where is the swelling located : ankle (R), ankle (L), calf (R), calf (L), tibia (R)
- Yes – Have you noticed any new fatigue , generalized and vague discomfort , diffuse (widespread) muscle aches or a change in your general well-being related to your consultation today
- Yes – Are your symptoms worse when lying down and alleviated while sitting up
History :
- Yes – Have you ever had fluid in your lungs
- Yes – Do you currently undergo dialysis
- Yes – Has high blood pressure or do you take medications to treat high blood pressure
- Yes – Have you ever had a heart attack or do you have angina (chest pain)
- Yes – Has heart failure
- Have you traveled out of the country in the last 4 weeks : N
<[assistant] |>
Noted.<[endofcontext]>
<[user] |>
For the patient described earlier...most likely diagnosis...
A) Pulmonary embolism
B) Possible NSTEMI/STE MI
C) Anaphylaxis
D) Acute pulmonary edema
E) Myocarditis
Answer :<[assistant] |>

back_10 – evidence share 0.0522

does not involve violence
D) I s not always hate speech , because it always involves violence .

Reply with only the letter (A, B, C, or D).

<[assistant] |>
A<[endofcontext]>
<[user] |>
Patient : 76-year -old Female
Chief complaint : Are your symptoms worse when lying down and alleviated while sitting up
Symptoms :
- Yes – Has pain somewhere , related to your reason for consulting
- Characterize your pain : tedious , sensitive , heavy
- Do you feel pain somewhere : lower chest , side of the chest (R), side of the chest (L), posterior chest wall (R), posterior chest wall (L)
- How intense is the pain : 9
- Does the pain radiate to another location : biceps (R), chin , jaw , shoulder (R), shoulder (L)
- How precisely is the pain located : 4
- How fast did the pain appear : 1
- Yes – Is experiencing short ness of breath or difficulty breathing in a significant way
- Yes – Has bouts of choking or short ness of breath that wake you up at night
- Yes – Has swelling in one or more areas of your body
- Where is the swelling located : ankle (R), ankle (L), calf (R), calf (L), tibia (R)
- Yes – Have you noticed any new fatigue , generalized and vague discomfort , diffuse (widespread) muscle aches or a change in your general well-being related to your consultation today
- Yes – Are your symptoms worse when lying down and alleviated while sitting up
History :
- Yes – Have you ever had fluid in your lungs
- Yes – Do you currently undergo dialysis
- Yes – Has high blood pressure or do you take medications to treat high blood pressure
- Yes – Have you ever had a heart attack or do you have angina (chest pain)
- Yes – Has heart failure
- Have you traveled out of the country in the last 4 weeks : N
<[assistant] |>
Noted.<[endofcontext]>
<[user] |>
The biblical account of the soul is at
– 649 tokens omitted –
<[assistant] |>
<[endofcontext]>
<[user] |>
As of 2017, the share of GDP spent on the military by Saudi Arabia is about
A) 1%
B) 3%
C) 6%
D) 10%
Reply with only the letter (A, B, C, or D).
<[assistant] |>
C<[endofcontext]>
<[user] |>
For the patient described earlier...most likely diagnosis...
A) Pulmonary embolism
B) Possible NSTEMI/STE MI
C) Anaphylaxis
D) Acute pulmonary edema
E) Myocarditis
Answer :<[assistant] |>

Figure 3: **Measured final-position attention, token by token** (OLMo-2, all-layer/head mean, log-scaled white→orange): windowed excerpts of the same probe’s transcript with its evidence local vs. displaced ten turns back. Solid underline marks the evidence span, dashed the current question, and elided stretches are labeled with their token counts. The evidence’s tint visibly fades when displaced (share 0.088 vs 0.052), and the hottest tokens in both arms are the chat template’s turn delimiters—the concentration whose causal role §4.5 establishes. Built from stored capture rows, not an illustration.

637 H Precedent details

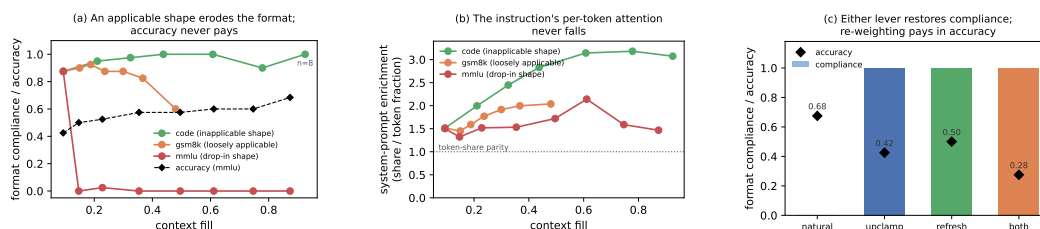


Figure 4: **Format erosion is a precedent contest, not attention starvation**, referenced from §4.4. (a) Compliance by fill for three filler streams whose demonstrated answer shapes differ only in applicability to the probe. mmlu accuracy (dashed) is unharmed through its total collapse. (b) The instruction’s per-token enrichment is flat-to-rising in every arm, collapse included. (c) At depth 42, clamping the system span’s share back up and restating the policy each restore compliance fully. Forced mass is paid for in accuracy (zero-sum), restatement is nearly free.

638 **The system span’s natural trajectory.** Measured in the clamp experiment’s own setup, the system
 639 span’s share as prior cases accumulate is 0.166 (0 cases), 0.081 (1), 0.060 (2), 0.037 (4), 0.028 (6),
 640 0.021 (8): an $8\times$ decline in eight turns from accumulation alone. Clamp ladders must be derived
 641 from the setup they are applied to. A share imported from a different context put two arms above
 642 natural, which were skipped rather than silently clamped upward.

643 **One contrary example overrides the instruction entirely.** With a single prior case whose answer
 644 exhibits the canaries, compliance is 3.00/3 at every clamp level down to share 0.021. With a single
 645 bare-letter answer it is 1.00/3 at every level including unclamped. 720 generations per arm, zero
 646 variance in reply length (8 characters in one arm, 1 in the other): the model reproduces the previous
 647 assistant turn’s format exactly. Both arms are pinned, one at ceiling because the format is available
 648 from the transcript and one at floor because the contrary example already won, and the clamp has
 649 nothing to move. Only the neutral condition (no assistant turn anywhere) exposes the instruction’s
 650 causal necessity reported in §4.4. A “no demonstration” arm whose prior turn answers with a bare
 651 letter is not an absence of demonstration but a counter-demonstration.

652 **Exemplar closure.** At depth 42, closing the demonstrated answer spans’ key columns at generation
 653 time (scale 0, §2.3): all-answer closure 0.000 compliant, dose-matched partial closure 0.000, filler-
 654 question closure 0.132 (0.100 in the committed re-run, both consistent with pure renormalization of
 655 the surviving spans, the system’s included), size-matched random-token closure 0.000. Accuracy
 656 moves within noise. Reading the exemplars at generation time is not what sustains the erosion.
 657 Figure 5 sets this against the competition rescue: the same intervention, opposite outcomes.

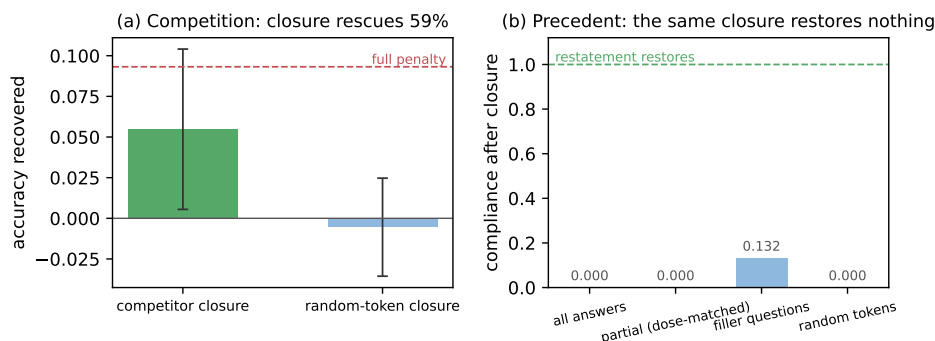


Figure 5: **One intervention, opposite outcomes.** The same scale-0 generation-time closure (a) recovers 56% of the competition penalty (paired $n=365$, bars are paired accuracy deltas with 95% CIs, dashed line marks the full in-run penalty), but (b) restores no compliance at depth 42 when applied to the demonstrated answer spans ($n=40$ /arm, every closure arm sits at or near the eroded floor while restating the instruction restores 1.000). Competition is generation-time reading. The precedent mode is already in place by prefill.

Probes. Probe 1 (instruction presence): trained at depth 0 to separate the format system prompt from a token-length-matched neutral twin. Transfer AUC is 1.000 at every mmlu depth 0 $\rightarrow$ 42 (permutation nulls 0.49–0.50, $p < 0.005$, $n = 80/76$). The only depth trend is concentration: mean-over-layers AUC 0.985 $\rightarrow$ 0.791, peak layer drifting L1–2 $\rightarrow$ L11–14. Probe 2 (upcoming compliance): within gsm8k’s mixed cells (same depth, fill, and filler, only the outcome differs, $n = 80$), LOO-AUC 0.875 at stack layer 22 (null 0.485, $p < 0.005$), crossing 0.8 from layer 18. Vector geometry: the mmlu and gsm8k mode vectors (deep minus shallow mean differences) have median cosine +0.746 across layers—two textually opposite demonstrated styles shift the residual stream in largely the same direction, with the caveat that generic deep-context components are common to both.

Steering and erasure. Install: the mean-difference vector at stack layer 21, applied at the final prefill position and every decode step at matched norm, destroys the instructed format in a demonstration-free context (0.000 vs 0.425 compliant under the norm-matched random control, natural 0.875), with replies reproducing the demonstrated registers verbatim. The collateral accuracy cost (0.175 vs 0.375) is consistent with deep-context components inside the mean-difference vector. A $3\times$ dose breaks generation for real and random alike—controls must be matched per dose. Erase: four strategies all null with clean controls—the mean-difference direction projected out at one layer and at eleven (the eleven-layer projection actively harms while its random control is harmless), and Probe 2’s own reader direction projected out at its own layer, both in-distribution and transferred. Iterative re-probing on the captured states shows the layer-22 linear code is rank- ≈ 2 (AUC 0.875 $\rightarrow$ 0.619 after one fitted projection, 0.505 after two), and the erase nulls do not reflect redundancy: the behavioral decision does not consume the linearly decodable component.

I Signature details

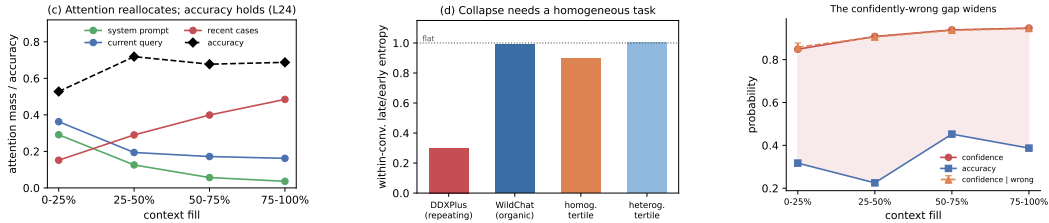


Figure 6: **The signatures track task homogeneity, and the residual hazard is calibration**, referenced from §4.6. (a) Attention reallocates monotonically off the system prompt and current query with fill while accuracy holds. (b) The entropy collapse appears only with a homogeneous repeating task, not on organic WildChat dialogue, and tracks homogeneity rather than length. (c) Confidence rises with fill on wrong answers as much as right, at flat accuracy.

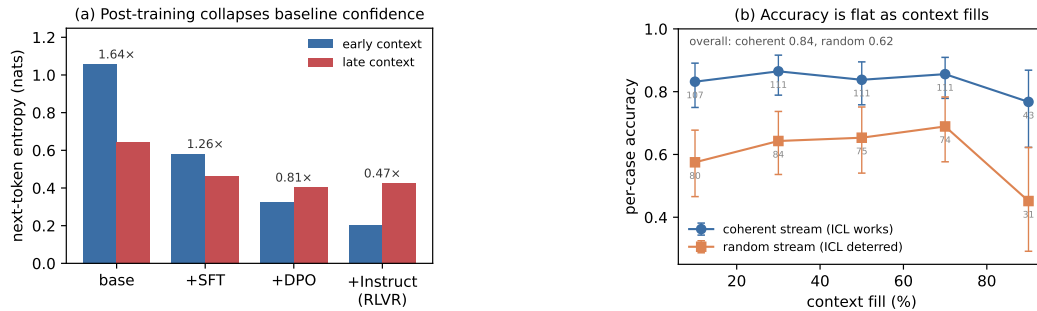


Figure 7: **Signature details.** (a) OLMo-2’s post-training chain collapses baseline entropy stage by stage. The within-context ratio falls across stages, so alignment installs a level shift, not a steeper slope. (b) Accuracy is flat with fill in both the coherent and the random-subject stream. The random stream is lower throughout (harder subjects), not sloped. Error bars are Wilson 95% intervals with per-bin n printed. The final bins are the smallest ($n = 31/43$) and their intervals overlap the preceding bins’, so the endpoint movement is not resolved at this n .

681 **Post-training chain.** Replaying one probe across OLMo-2-7B’s base→SFT→DPO→Instruct chain
682 (identical cases and orders) shows early-context entropy collapsing monotonically, $1.06 \rightarrow 0.58 \rightarrow$
683 $0.33 \rightarrow 0.20$, with the largest single drop at base→SFT (Fig. 7a). The “collapses as context fills”
684 framing is mostly a base-model ICL effect plus a floor: the within-context ratio falls across stages
685 ($1.64 \rightarrow 0.47$), because aligned models start near an entropy floor.

686 **WildChat.** Median within-conversation $\text{corr}(\text{entropy}, \text{depth})$ is -0.02 on organic dialogue (150
687 conversations, 2,039 boundaries). The homogeneity partition is a separate 400-conversation run:
688 tertiles of inter-turn TF-IDF similarity give late/early ratios 0.897 (homogeneous) vs 1.001 (het-
689 erogeneous), $n=134$ per tertile. Homogeneity correlates with the entropy slope (Pearson -0.141
690 at $p=0.005$, Spearman -0.103 at $p=0.039$, partial $r = -0.151$ controlling for length) and with
691 conversation length ($+0.155$: homogeneous conversations are longer, and length works against the
692 effect). The tertile medians are reported without intervals.

693 **A proposed mechanism that fails causally.** Clamping the boundary-detector SAE feature (Gemma
694 Scope F90871, Lieberum et al., 2024) back to its clean level on held-out fatigued cases moves entropy
695 the wrong way ($0.442 \rightarrow 0.465$ vs clean 0.293), leaves accuracy flat, and a random feature of similar
696 magnitude perturbs entropy at least as much.

697 **Attention and errors point the other way.** Conditioning on fill, the cases OLMo-2 answers wrong
698 have higher current-query attention than the ones it gets right (pooled $\Delta = +0.045 [-0.003, +0.097]$,
699 $n=115$). The interval touches zero, but the sign is positive in every fill bin and at every probed layer,
700 an observational pattern rather than an established one. Errors co-occur with fixating on the query
701 while correct answers spread attention onto the accumulated cases—the per-case fingerprint of the
702 in-context learning that keeps accuracy up, with the “rot” and the within-context predictor of a correct
703 answer pointing in opposite directions.

704 J Cross-family replication: Qwen2.5-7B-Instruct

705 The causal program was re-run on Qwen2.5-7B-Instruct (seed 42, 4k window, same drivers, panel
706 constructions, and drop rules as the committed OLMo runs). The window matches the OLMo runs,
707 and the cross-family contrasts below, the competition divergence included, are not window effects.
708 Share readouts are all-layer pooled except the distance sweep’s, which is layer-24-indexed like
709 OLMo’s. Table 10 is the summary; the per-experiment detail follows it. Nine of the paper’s claims
710 reproduce, two reproduce in part, and the three that make up the competition result do not.

Table 10: **Every claim, in both families.** Accuracy contrasts in percentage points, compliance as a fraction. “partial” marks a claim that holds in direction but not in degree; “inverted” marks a measurement whose sign differs with disjoint intervals. Two rows quote the statistic each run recorded rather than a common one: the accumulation row is OLMo’s fill correlation against Qwen’s fill slope (both over random and coherent streams, all n.s.), and the system-clamp row is OLMo’s compliance collapse against the share at which each of Qwen’s two canary duties fails. Both families run at seed 42 in a 4k window on the same panels, so no row is a window effect.

claim	OLMo-2-7B	Qwen2.5-7B	holds
accumulation costs no accuracy	$r = -0.02, +0.03$	slope $-0.06, -0.09$	yes
displacement ladder is monotone	$0.464 \rightarrow 0.276$	$0.630 \rightarrow 0.469$	yes
and saturates by $k \approx 10$	0.250 then 0.276	0.505 then 0.469	partial
mass removal reproduces the gap	91%	107%	yes
the dose-response is graded	$0.473 \rightarrow 0.275$	$0.667 \rightarrow 0.380$	yes
with no threshold in it	max step 0.053	no knee	yes
clamping the system span kills compliance	$0.99 \rightarrow 0.03$	at shares $0.12, 0.09$	yes
erosion tracks applicability	$1.00/0.60/0.00$	$1.00/1.00/0.00$	yes
and is total	$0.875 \rightarrow 0.000$	ANSWER: only	partial
restating the instruction reverses it	1.000	1.000	yes
re-weighting it reverses it	1.000	0.733	partial
the demonstrated answers are read	$+1.28$	$+0.87$ to $+1.49$	yes
near-duplicate context costs accuracy	$+8.5$	$+3.0$, n.s.	no
closing the competitors rescues it	$+6.0$ (56%)	$+1.9$, n.s.	no
the evidence’s share under <code>near_dup</code>	$+0.0003$, n.s.	$+0.0071$	inverted

Two qualifications the table cannot carry. The cross-family difference-in-differences on competition (OLMo’s paired penalty minus Qwen’s) is $+5.5$ points $[-1.5, +12.5]$, suggestive and not significant, so the divergence rests on the closure null and the inverted share rather than on a direct interaction test. And Qwen’s low-competition arms sit ≈ 10 points above its own local arm, because same-task context practices the probe’s task where MMLU filler does not; cross-family comparisons of arm levels inherit that practice benefit, and only within-family gaps are interpreted.

Table 11: **Counterfactual patching of the format-instruction state** (Qwen2.5-7B, code cell), referenced from §4.5. Full-transcript row $n=39$, bisection rows $n=23$. $\Delta\Delta = \Delta\ell_{\text{patch}} - \Delta\ell_{\text{self}}$: the shift of the instructed-formats log-probability contrast $\Delta\ell$ against a self-patch baseline.

patched positions	$\Delta\Delta$	share of full
full context, all layers	-2.488	100%
template glue (<code>im_end/im_start</code>)	-1.540	62%
assistant turns	-0.287	12%
user turns	$+0.165$	—
last 1/2/4 turns	$\leq 0.075 $	~ 0
size-matched random positions	$+0.016$	—
template glue $\times$ layers 7–13	-1.051	42%

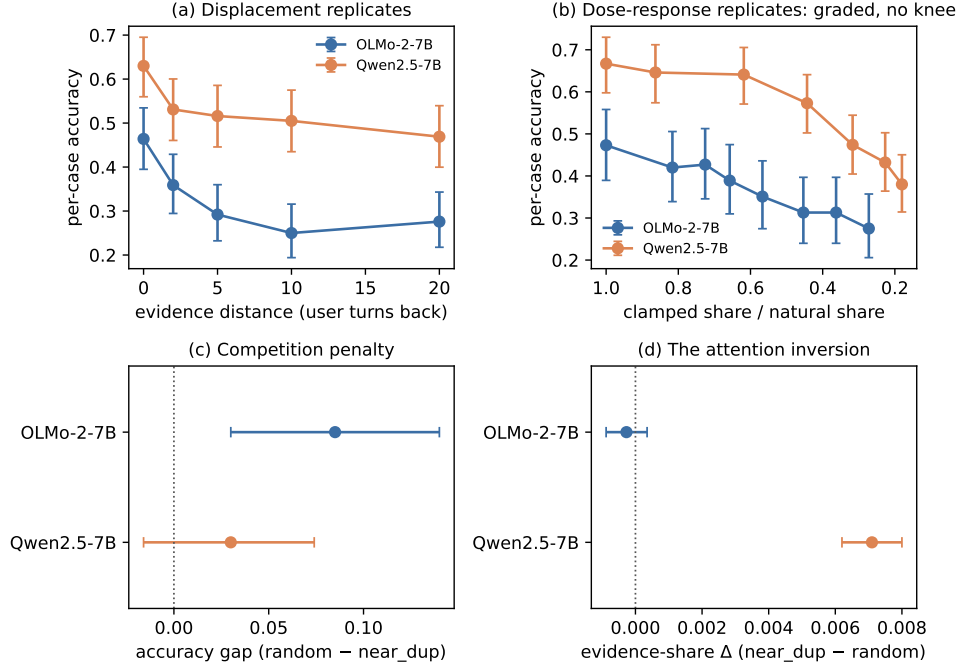


Figure 8: **The causal program replicates on Qwen2.5-7B, except competition.** (a) The distance ladder at matched fill ($n=192/\text{arm}$ in both families). (b) The share $\rightarrow$ accuracy dose-response, x as a fraction of each family’s natural share (OLMo layer-24-indexed with common subset $n=131$, Qwen all-layer with $n=181-192$): graded with no knee in both. (c) The competition contrast is significant on OLMo, not detected on Qwen. (d) The evidence’s attention under near-duplicate context: flat on OLMo, rising on Qwen, with disjoint intervals—the family divergence. Whiskers in (a, b) are per-point Wilson 95% intervals. Panels (c, d) show the paired bootstrap CIs quoted in the text.

717 **Displacement.** The ladder reproduces fully monotone: $0.630 \rightarrow 0.531 \rightarrow 0.516 \rightarrow 0.505 \rightarrow$
718 0.469 ($n=192$ per arm, 0 overflow skips), every paired gap significant (back_2 +9.9 points
719 $[+5.2, +14.6]$ through back_20 +16.1 $[+9.4, +22.9]$). In the joint fit distance carries the coefficient
720 $(-0.0061 [-0.0102, -0.0016])$ and fill is significant only with the opposite sign (+0.330)—byte-
721 identical fill across arms cannot carry the ladder. Qwen answers with a bare letter (0.0% unparsed),
722 and the ladder needs no parsed-only robustness arm: the truncation caveat attached to the OLMo
723 ladder does not arise on this family.

724 **Mass sufficiency and dose-response.** Clamping local evidence to back_20’s share repro-
725 duces 107% of the distance gap (mass alone +16.7 points $[+10.4, +22.9]$ against a gap of
726 +15.6 $[+8.9, +22.4]$). The clamped arm is indistinguishable from back_20 (residual -1.0 points
727 $[-5.7, +3.7]$). The share sweep is graded with no knee, still falling below the deepest natural share
728 (0.667 at natural to 0.380 at share 0.0070), and its causal slope, +10.4 $[+7.6, +13.6]$ accuracy per
729 unit share, matches the observational fit of correctness on share (+10.7 $[+3.5, +17.6]$) from a sep-
730 arate design (the observational fit is on the distance sweep’s layer-24 shares, the causal slope on
731 all-layer shares, and the agreement is suggestive rather than unit-exact).

732 **Accumulation and the system clamp.** The random-subject stream is flat across its 4k window (fill
733 slope $-0.057 [-0.238, +0.135]$, with the coherent stream at -0.090 , also n.s.), and the adherence
734 canaries do not decay. Clamping the system span’s share collapses compliance at graded dose with
735 a duty ordering, the reply’s separately-graded obligations failing one at a time: reply-prefix canary
736 first (at share 0.12), suffix next (0.09), the forbidden-phrase rule never (Fig. 9). Meanwhile accuracy
737 rises under the clamp ($-0.09 [-0.16, -0.03]$ at the deepest level), the zero-sum reallocation that was
738 directionally present but n.s. on OLMo.

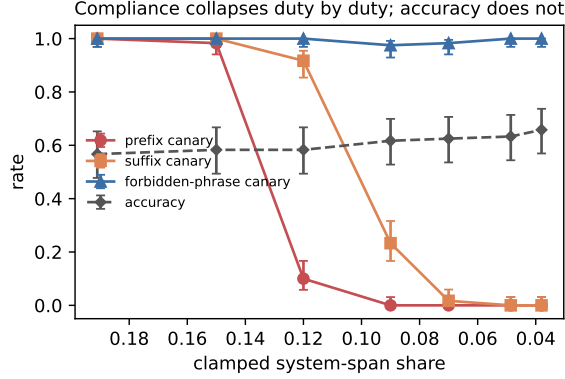


Figure 9: **Qwen’s system clamp collapses compliance duty by duty.** Neutral-context clamp ($n=120/\text{level}$): the reply-prefix canary dies between shares 0.15 and 0.12, the suffix between 0.12 and 0.09, the forbidden-phrase rule never. Accuracy is flat-to-rising throughout. Whiskers are Wilson 95% intervals.

Precedent. The applicability ordering confirms: `mm1u` compliance $1.000 \rightarrow 0.000$ within three filler turns while `gsm8k` and `code` hold 1.000 through fill 0.84–0.92, accuracy flat throughout (overflow skips thin code to $n=17$ at its deepest fill). The demonstrated-answer reading signature reproduces (answer-over-question enrichment $+0.87$ to $+1.49$ in `mm1u` at every depth, negative in `code`—the erosion ordering). Two results differ. Erosion is staged rather than total: the ANSWER: marker dies immediately while the SUPPORTING duty survives to mid-depth, the same duty ordering the clamp produces causally. Restatement beats re-weighting: at depth 42, refreshing the instruction restores 1.000 while upclamping the system span restores only 0.733 (both at no accuracy cost). One dissociation does not carry over: Qwen’s compliant arms sit at or above the share where its own clamp collapses the prefix duty and its eroding arm below it, and Qwen’s erosion arms do not by themselves exclude a mass account the way OLMo’s code arm (full compliance at half `mm1u`’s share) did.

Competition: the family divergence. The three competition rows of Table 10 are one result: on the same $n=365$ panel the penalty is not detected, the competitor closure that rescues OLMo does nothing, and the evidence’s share moves the other way with disjoint intervals. Qwen defends the evidence under confusability rather than merely not paying for it.

A masked-channel account is available. By Qwen’s own dose-response slope the share it gains under `near_dup` should be worth ≈ 7 points of accuracy, yet the arm underperforms that prediction by ≈ 10 points, close to OLMo’s penalty. On that reading Qwen pays the same cost and hides it behind a compensating attention gain. The account stacks coefficients estimated in different designs, and we mark it interpretive. What the closure null does establish either way is the channel: whatever the shortfall is, it is not carried by the competitor-mention tokens that carry OLMo’s cost.

J.1 Counterfactual instruction patching and the carrier bisection

This is the instrument behind §4.5’s token-level localization.

The instrument. Build two transcripts that are identical in filler and probe and differ only in the system instruction: T_A carries an ANSWER: prefix format, T_B a one-line JSON format. Fix one reply in each format, y_A and y_B , and score a transcript by how much it prefers one over the other under teacher forcing,

$$\Delta\ell(T) = \log P(y_A | T) - \log P(y_B | T).$$

$\Delta\ell$ is large when the model is disposed to answer in format A and small when it is disposed to answer in format B , so it reads out the disposition §4.5 calls the mode without requiring the model to emit anything.

A difference of log-probabilities between two fixed continuations is the readout Wang et al. (2023) established for circuit-level patching, and Zhang and Nanda (2024) survey it against the probability- and accuracy-based alternatives. Two of its properties decide the choice here. It is linear in the logits,

so a patch that moves the contrast by a given amount moves it by that amount wherever the model sits on its confidence curve, which a probability readout does not do once either format saturates. And it cancels any shift common to both continuations: the deep code cell runs with the system span closed, which depresses the likelihood of every well-formed reply at once, and a metric scoring one format alone would register that as an effect. Only the contrast between the two survives it.

Now move the state. Write $h_P(T)$ for the hidden states captured from T at a set of token positions P , and $T \leftarrow h$ for a forward pass on T whose prefill activations at those positions are overwritten with h . The counterfactual patch transplants the donor’s states into the recipient at P ; everything else about the recipient is untouched. This is activation patching in the causal-mediation form introduced by Vig et al. (2020) and used for factual recall by Meng et al. (2022), with the instruction rather than a fact as the mediated variable. All cells run in the $A \rightarrow B$ direction, A donating and B receiving, in the deep code cell at depth 15 with the system span closed. The full-transcript patch is the stage-1 run at $n=39$ and the bisection cells are $n=23$ after one overflow skip.

The estimand. The quantity in Table 11 is the interaction contrast

$$\Delta\Delta(P) = \underbrace{\Delta\ell(T_B \leftarrow h_P(T_A))}_{\text{patched with the donor's states}} - \underbrace{\Delta\ell(T_B \leftarrow h_P(T_B))}_{\text{patched with its own states}},$$

a difference-in-differences in which both arms undergo the identical capture-and-splice procedure at the identical positions, and the only thing that differs is whose states were spliced in. A large $|\Delta\Delta(P)|$ means the positions in P carry the format instruction’s effect; a $\Delta\Delta(P)$ near zero means they do not. The “share of full” column normalizes each cell against the whole context, $\Delta\Delta(P) / \Delta\Delta(\text{all positions})$.

The baseline is the self-patch rather than an unpatched run, and that choice is what makes the measurement work. Donor states are captured with the system span open, while a pure run forms its states with that span closed, so $\Delta\ell(T_B)$ and $\Delta\ell(T_B \leftarrow h_P(T_B))$ differ by a capture-openness term that has nothing to do with donor identity. In preflight an unpatched baseline confounded exactly that term with the effect of interest, and the unrelated-fact control moved as much as the counterfactual patch; against the self-patch baseline the same control is null. Identification then needs only additivity, that capture-openness shifts both arms’ contrasts by the same amount, which is what sharing one procedure across arms is designed to buy.

Where the state lives. The full-transcript patch moves the contrast by $\Delta\Delta = -2.488$. Bisecting by position, every content subset is near null. The context’s assistant turns carry -0.287 , its user turns $+0.165$, and its last 1/2/4 turns at most $[0.075]$. The chat template’s turn delimiters alone carry -1.540 , 62% of the full effect, against a size-matched random-position control of $+0.016$. Those delimiters contain no instruction text and no demonstrated answer. Bisecting by depth instead, layer-block patches at all positions give -1.16 (layers 0–6), -1.59 (7–13), -0.45 (14–20), and -0.11 (21–27). The blocks overlap in what they carry, so their sum exceeds the full effect. Crossing the two, glue positions restricted to layers 7–13 still carry -1.051 (42%) while touching no content token and no layer above 13. Earlier waves that searched inside the turns, by role and by recency, came back null: the state rides between the turns rather than in them.

Specificity control. The control throughout is an unrelated-fact patch, the same instruction with one irrelevant clause changed, token-matched. It stays at $\leq |0.11|$ in every quoted cell except two, and both of those have counterfactual effects that are themselves near null: the last-2-turns cell at $+0.26$ and the random-position cell at -0.11 .

Two caveats on the readout. Under closure Qwen emits no canonical format in free generation, in none of the 90 graded replies across both patch arms. The estimand lives in the teacher-forced contrast alone, with no generation-level corroboration in this cell. Closure also drops the recipient’s system-span identity out, which makes the two patch directions exact mirror images ($\Delta\Delta_{B \rightarrow A} = -\Delta\Delta_{A \rightarrow B}$), so the quoted direction is the one independent measurement. The full effect is also sign-inverted relative to the donor’s instruction, in that A-donor states depress the A-format productions. The localization claim does not depend on that. The bisection asks where the state travels, not which way it pushes.